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PROFILE

Robotics & AI Engineer with expertise in ROS2, autonomous navigation, computer vision, and Human-Robot Interaction. Recently completed an M.Sc. in Electromobility ACES at Friedrich-Alexander-Universität Erlangen-Nürnberg, with experience developing intelligent robotic systems, sign language enabled humanoid interactions, and assistive robots for healthcare applications. Experienced in both industrial software development and advanced robotics research.

TECHNICAL SKILLS

Programming: Python, C++, C#

Robotics & Simulation : ROS2 (Humble), ROS2 Control, Gazebo, SLAM, Nav2

AI & Computer Vision: OpenCV, TensorFlow, Keras, Deep Learning, Computer Vision

Tools & Systems: MySQL, SSMS, VS Code, Git, Linux

CAD: Autodesk Inventor, SolidWorks

Testing: xUnit, NUnit (in C#) **Language Skills:** English (C1), German (B1.1)

PROFESSIONAL EXPERIENCE

Neura Robotics

March 2025 - October 2025

Neura Robotics Challenge (Top 5 Europe-wide Finalists)

Germany

- Proposed a novel solution to implement on the humanoid robot to bridge the communication gap between sign language users and non-signers.
- Compiled a dataset of 3000+ sign language samples representing hand poses due to the lack of truly open-source data in this field.
- A RandomForest classifier and an LSTM were successfully trained to recognize these signs.
- Achieved an accuracy of 85% on a custom dataset using the RandomForest classifier and 91% using the LSTM.
- Implemented forward kinematics and programmed the robot to respond to a set of predefined questions in sign and spoken language.
- Demonstrated the potential for real-time communication using humanoid robots to improve inclusivity. (Featured in [FAU News](#))

Deloitte USI

August 2021 - March 2023

Software Analyst

Hyderabad, IN

Test Engineer - General Motors Financial

- Designed and executed test cases for migration of the enterprise financial systems to cloud-based infrastructure.
- Validated application logic, data integrity and the performance of the systems during migration of client-facing financial workflows.
- Collaborated with development teams to identify more than 20 breaking defects per subsystem to increase the workflow stability before deployment.
- Contributed to CI/CD pipeline testing, enabling smoother release cycles and reducing build failures during deployment phase.

Automation Developer - CommonSpirit Health

- Developed and maintained over 10 automation scripts using C# and Summit Scripting for healthcare billing and EMR processing.
- Led the development of core automation module for markets generating close to 22% of the client revenue, reducing manual intervention and human errors.
- Reduced the insurance claims processing times from days to less than 6 hours through automation workflow optimization.
- Implemented secure data handling techniques and encryption to handle sensitive patient data in compliance with healthcare standards.
- Improved system efficiency by scheduling tasks and automating the execution to enable a fully automated execution.

Ashok Leyland - Engine Division

May 2019 - June 2019

Intern (Quality Division)

Hosur, IN

- Analyzed factors affecting liner projection in IC engine blocks which led to quality issues and liner failures.
- Collected sensor data from over 500 Engines and compared it with manual measurements during the manufacturing process to identify potential root causes of catastrophic engine gasket failure.
- Tested sensors for improper calibration and measured environmental factors affecting the measurement processes.
- Used data-driven analysis to isolate manufacturing faults and improve reliability in tight tolerance machining process.
- Identified and reported factors affecting processes such as imperfections in machining tool alignment defects, probe errors, temperature and load correlation to implement necessary solutions.

ROBOTICS AND AI PROJECTS

20 DoF Simple Kinematics Robotic Hand

October 2025 - April 2026

Friedrich-Alexander-Universität Erlangen-Nürnberg

- Redesigned robotic hand with actuator redundancy (20 DoF) for smoother human-like movement.
- Redesigned kinematics to prioritize human-like motion over focus on grasp capabilities.
- Designed actuator placement within finger links to simplify kinematics.
- Developed low-power motor control using angular position sensors.

Social Robot for Patients with Cognitive Impairments

February 2025 - July 2025

Friedrich-Alexander-Universität Erlangen-Nürnberg

- Developed a SLAM based social robot using the TurtleBot platform for dementia patients, achieving a success rate of 96% with 60% autonomous recovery
- Implemented navigation assistance for patients to traverse familiar indoor environments such as in care facilities to help with spatial disorientation.
- Investigated the walking speed of patients suffering from chronic disabilities to set the bot speed at 0.3 m/s.
- Implemented SLAM and Nav2 stack in ROS2 using Gazebo simulation, enabling autonomous navigation in indoor environments.
- Added additional functionality to recognize the faces of close friends and family to help the patient remember during memory lapses.
- Integrated computer vision-based face recognition using OpenCV and deep learning models

Graph Convolutional Networks for Physical Property Prediction

October 2020 - March 2021

National Institute of Technology Karnataka

- Analyzed the relationship between molecular structure of certain chemicals and their macro physical properties.
- Extracted the molecule composition along with its structural information as a Simplified Molecular Input Line Entry System (SMILES string).
- Trained a modified GCN model on a training dataset and validated the model using a separate validation dataset.
- Improved model performance test scores by 29.59% using the GraphConv method compared to the baseline model.
- Optimized computational resources by reducing runtime by 17%, while enhancing accuracy and F1-score, and lowering MSE.

EDUCATION

Master of Science in Electromobilität ACES

April 2023 - May 2026

Friedrich-Alexander-Universität Erlangen-Nürnberg

Bachelor of Technology in Mechanical Engineering

June 2017 - June 2021

National Institute of Technology Karnataka, Surathkal

POSITIONS OF RESPONSIBILITY

ISTE CLUTCH Team Lead ISTE SIG Head - NIT Karnataka

June 2020 - May 2021

- Headed the Indian Society for Technical Education (ISTE) Special Interest Group named 'Clutch' overseeing operations and project initiatives.
- Organized 2 technical projects (Modular E-Skateboard and Manufacturing of Glass Fiber Reinforced Resin Based Composite) at the college's year end technical fest as a lead in charge of the projects.